

CHETAN PATIL

B.Tech Aeronautical Engineering · Final Year (7th Semester) · Acharya Institute of Technology, Bengaluru

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OBJECTIVE

Final-year Aeronautical Engineering student with hands-on experience in UAV systems, aircraft design, propulsion sizing, structural analysis, and CAD modelling. Passionate about applying engineering fundamentals to real-world aerospace challenges. Seeking internship or project opportunities in aeronautics, UAV development, or aerospace research where I can contribute and grow.

EDUCATION

B.Tech – Aeronautical Engineering · Acharya Institute of Technology, Bengaluru 2023 – Present

- CGPA: 7.79 / 10.0 (Semesters 1–5) · Entering Semester 7 (Final Year) · Expected Graduation: 2027
- Relevant Coursework: Aerodynamics, Aircraft Structures, Propulsion Systems, Flight Mechanics, Stability & Control

Program / Level	Institution	Score	Year
Class XII – Pre-University College	Empower PU College, Dharwad	80.0%	2023
Class X – SSLC	Mahatma Gandhi High School, Yadgir	73.33%	2021

TECHNICAL SKILLS

Aeronautics	Flight Mechanics, Aerodynamics, Stability & Control, UAV Systems, Aircraft Structures, Propulsion
CAD / Design	Onshape, CATIA / SolidWorks — 3D modelling, assembly design, airframe optimisation
Computation	MATLAB (numerical modelling, curve-fitting), Python (data processing, scripting, automation)
Design & Build	RC aircraft & drone fabrication, payload integration, propulsion sizing, PID tuning
Tools & OS	MS Office (Word, Excel, PowerPoint), Linux & command-line, Git (basic)
Languages	English (fluent) · Hindi (proficient) · Kannada (proficient)

ACADEMIC & TECHNICAL PROJECTS

Agricultural UAV Prototype — System Integration & Propulsion · Smart India Hackathon · National Level · Team Lead 2023–24

- Led aerodynamics and propulsion sub-team of 6; optimised airframe to hit payload-to-weight ratio target within a 72-hour build constraint.
- Integrated ESC, flight controller, and spray payload within structural airframe; managed vibration isolation and thermal load on electronics.
- Delivered a flight-ready prototype — one of the top entries at the SIH institutional round.

RC Fixed-Wing Aircraft — Design, Build & Fly · College Aero Team 2024

- Designed wing geometry, sized propulsion for target thrust-to-weight ratio, and performed CG and stability margin analysis before first flight.

- Applied coursework stability-margin calculations to reduce trim iterations by ~40% between builds — achieved stable, controlled first flight.

RC Drone & RC Robocar — Build & Compete · Shristi Tech Event · Participation Certificate 2024

- Independently designed and built an RC drone from scratch — configured flight controller, tuned PID parameters, and achieved stable hover on event day.
- Simultaneously engineered RC robocar with motor-driver logic and steering control — both systems competed successfully under competition time constraints.

Oblique Impact Dynamics — Experimental & Analytical Study · 5th Semester Mini Project 2025

- Ran 50+ impact trials across 5 incidence angles and 3 surface types; validated theoretical friction-viscoelastic model to < 8% error.
- Results applicable to aircraft tyre touchdown, landing gear design, and turbine blade erosion behaviour.
- Processed raw data in Python and delivered a full technical report with uncertainty quantification and aerospace application mapping.

Soapbox Car — Structural Design & Fabrication · Inter-College Tech Event 2025

- Engineered load-bearing chassis under strict weight budget applying material selection and joint analysis under dynamic loading.
- Vehicle completed full race course — ranked top-3 in structural integrity judging at the event.

CERTIFICATIONS

Aircraft Design · NPTEL · 12-Week Course 2024–25

- Completed 12-week NPTEL course; cleared offline proctored certification exam — one of fewer than 30% of enrolled students to earn the certificate.

Flight Mechanics Basics · Coursera

Aircraft Fluid Dynamics · Coursera

EXTRACURRICULAR & VOLUNTEER ACTIVITIES

- Active member — Aeromodelling & Robotics Club; built and flew RC aircraft and competed in soapbox car race (Tech Event 2025).
- Participated in college-level football, basketball, and cricket.
- Volunteer — Shrishti Tech Event 2024, Cultural Fest 2024, Tech Fest 2024: managed participant coordination, event logistics, and on-ground support.

DOMAINS OF INTEREST

UAV Systems · Aircraft Design & Structures · Aerospace Propulsion · Stability & Control · Automation · Cybersecurity (Aerospace Applications)